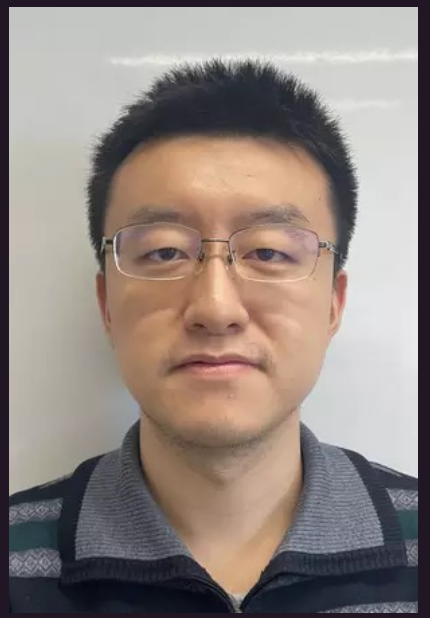
Introduction and Course Overview

Chirag Raman
Assistant Professor
Department of Intelligent Systems, TUDelft

Generative Modeling DSAIT4030, 21/04/2026

Welcome!

Responsible instructor	Lecturer, TA	Lecturer, TA	Lecturer, TA	Lecturer, TA	Lecturer, TA
					
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Welcome!

TA



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Lara Santos Bastos

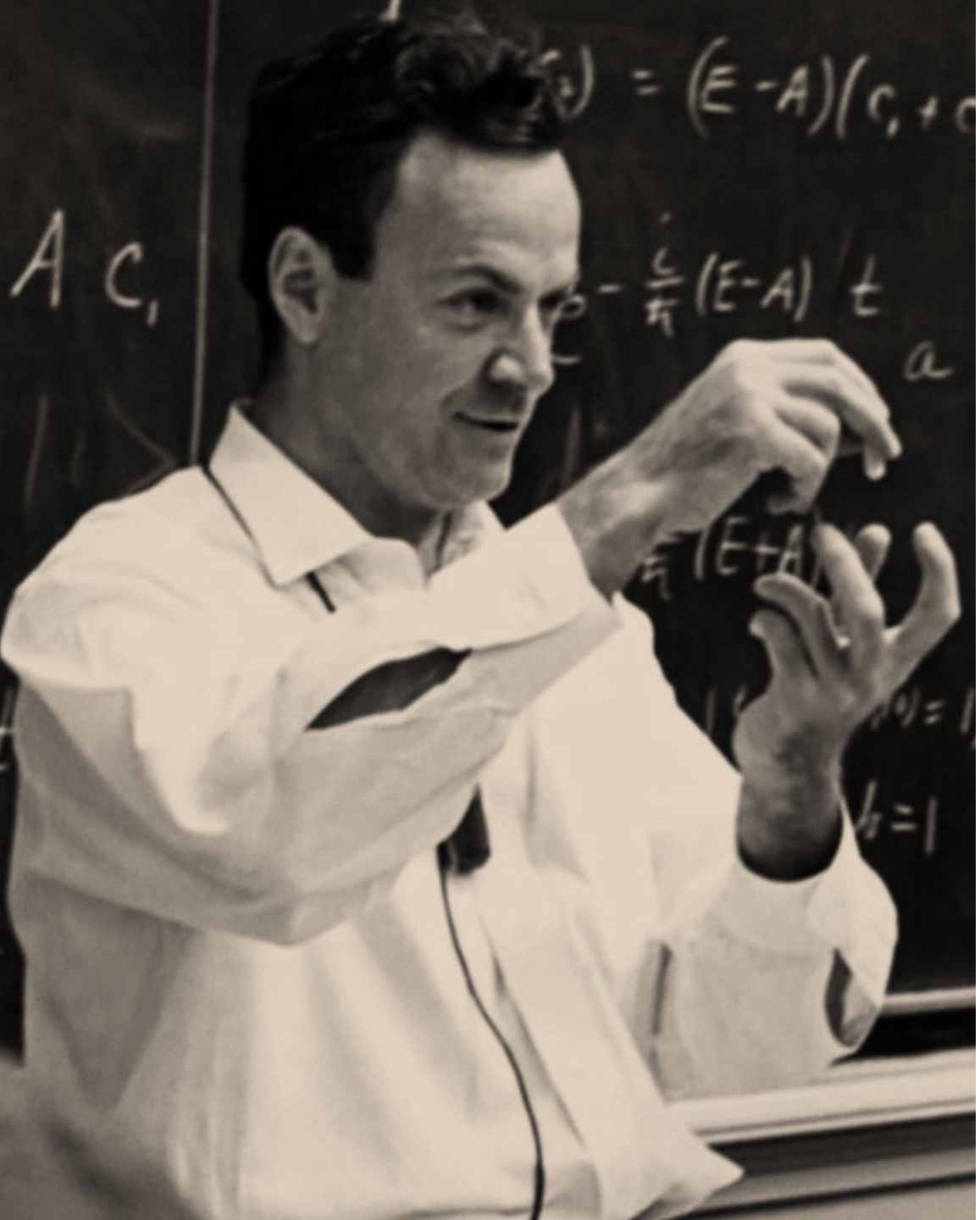
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TA



Arthur Mercier

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"What I Cannot Create, I Do Not Understand."

Richard Feynman

Outline

Goal: What is Generative AI? Guiding intuitions?

By the end of this lecture, you will be able to

- Outline a brief history of modern AI and trace the development of several techniques that are relevant currently
- Identify the goals of AI, Machine Learning, and Deep Generative Modeling
- Discuss Frequentist and Bayesian interpretations of probability
- Contrast discriminative and generative modeling approaches
- Identify the core ideas underpinning different families of generative modeling techniques

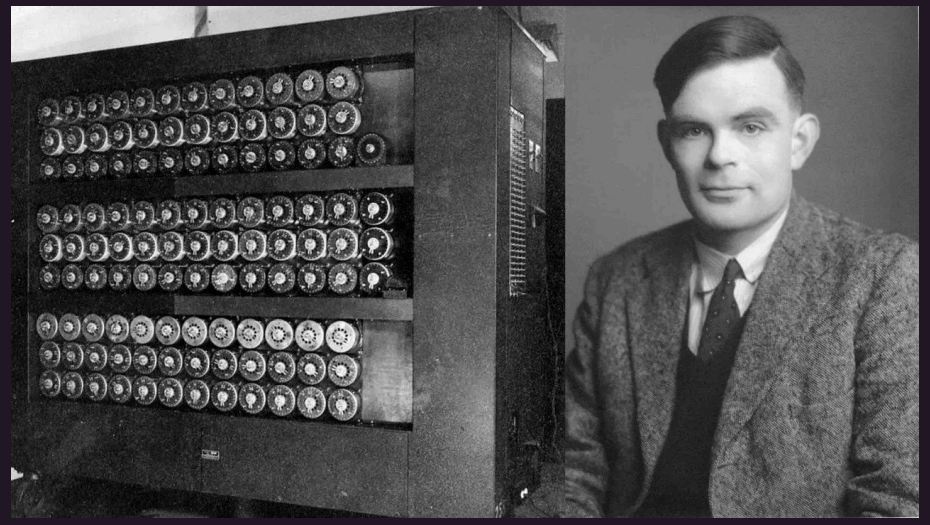


Modern History of AI

1936
1950

Alan Turing

The Turing Machine and
The Turing Test



1956

McCarthy et al.

Coin the term
“Artificial Intelligence”



1966

ELIZA Chatbot

Passes the Turing test

```
Welcome to
                EEEEEE  LL      IIII   ZZZZZZ   AAAAA
                EE      LL       II       ZZ   AA   AA
                EEEEE   LL       II     ZZZ    AAAAAAA
                EE      LL       II    ZZ      AA   AA
                EEEEEE  LLLLLL  IIII ZZZZZZ    AA   AA

  Eliza is a mock Rogerian psychotherapist.
  The original program was described by Joseph Weizenbaum in 1966.
  This implementation by Norbert Landsteiner 2005.


ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

1974-1980

1st AI Winter

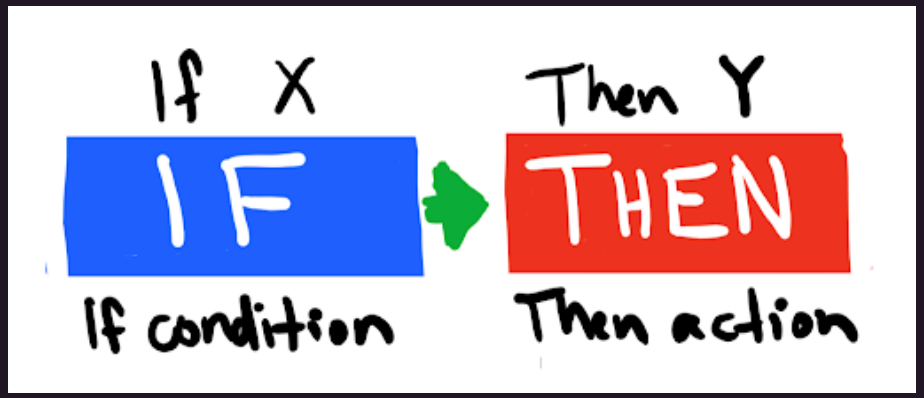
General funding cutbacks



1980s

Rise of Expert Systems

Representing human decisions in if-then form
“AI Renaissance”



1987-1993

2nd AI Winter

Limitations of if-then reasoning

CNNs (LeCun, 1989)



1997

Deep Blue

Defeats chess champion

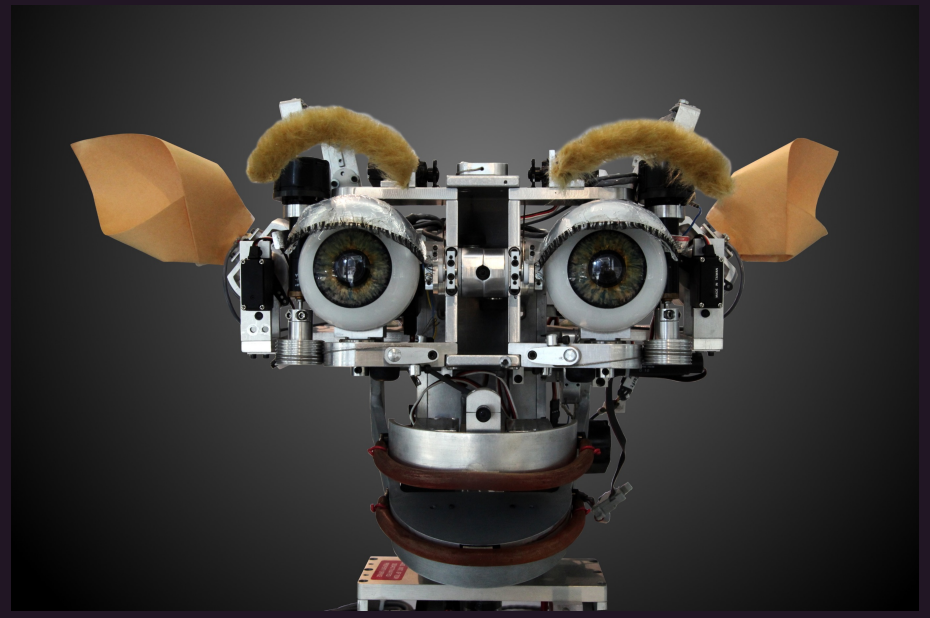
RNNs (Hochreiter)



Early 2000s

Robots

Kismet, AiBO, Roomba, Stanley



2011

Siri, IBM Watson

Apple integrates virtual assistant into iPhone

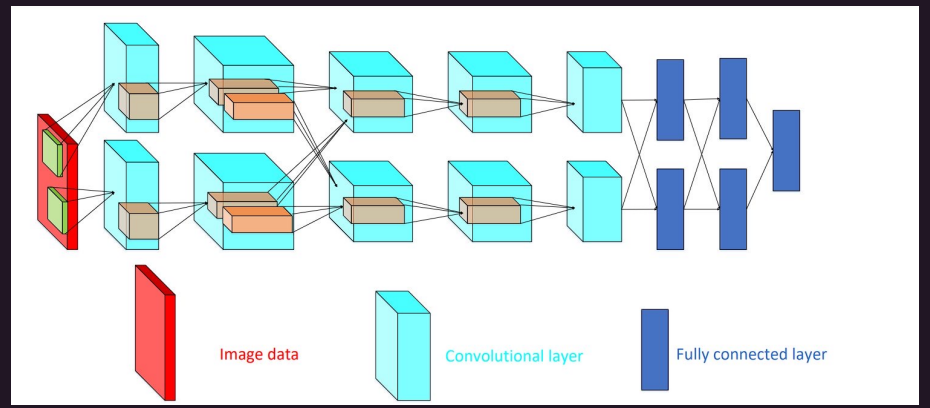
Watson wins Jeopardy



2012

AlexNet DCNN

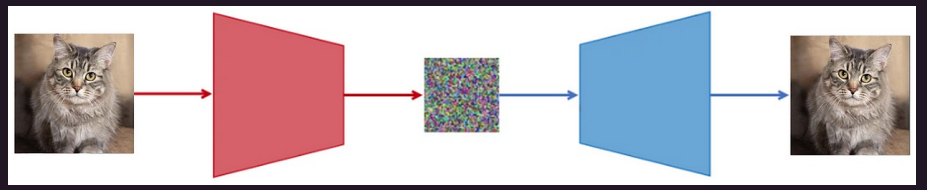
Wins ImageNet challenge for visual recognition



2012-2015

RL, VAE, GAN, Diffusion

Rapid development of deep learning techniques



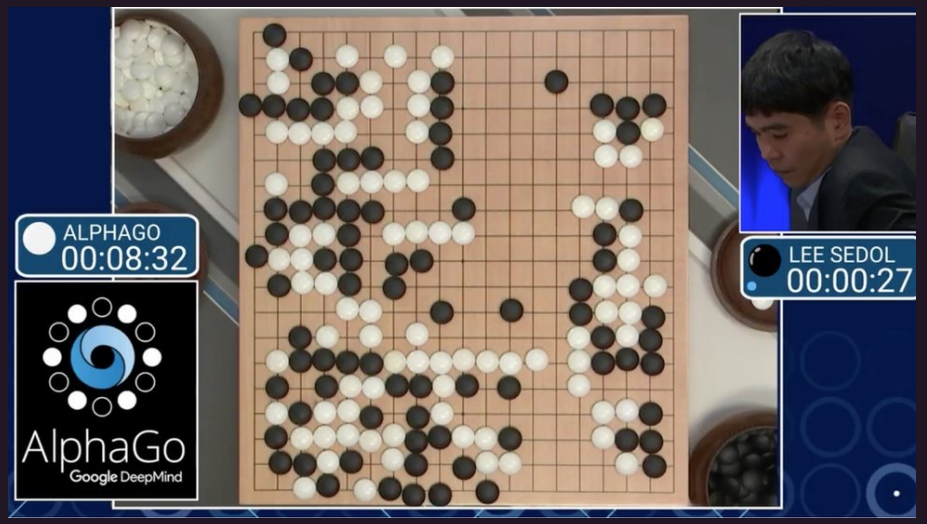


2016

AlphaGo, Uber

AI beats top Go champion

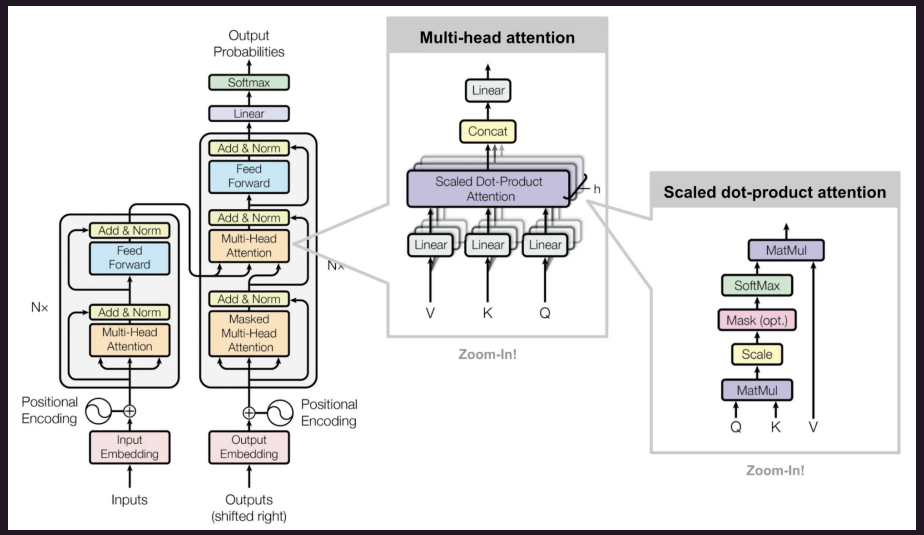
Uber pilots self-driving car program in Pittsburgh



2017

Transformers

“Attention is all you need”, inspiring subsequent research into LLMs

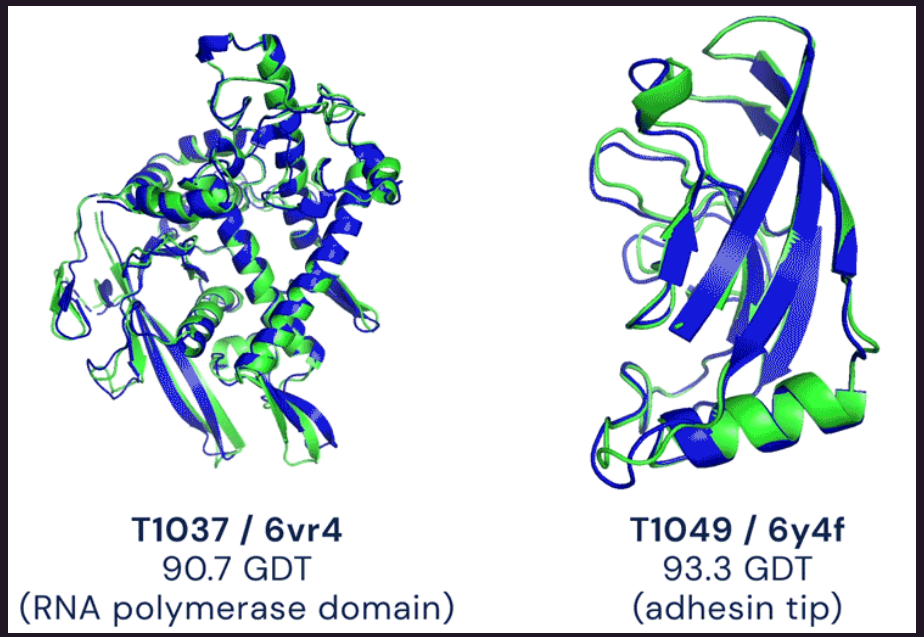


2020

GPT, AlphaFold

OpenAI releases GPT-3 with 175B parameters

DeepMind’s AlphaFold2 wins the CASP competition for protein structure prediction



2021-2022

Dall-E, ChatGPT

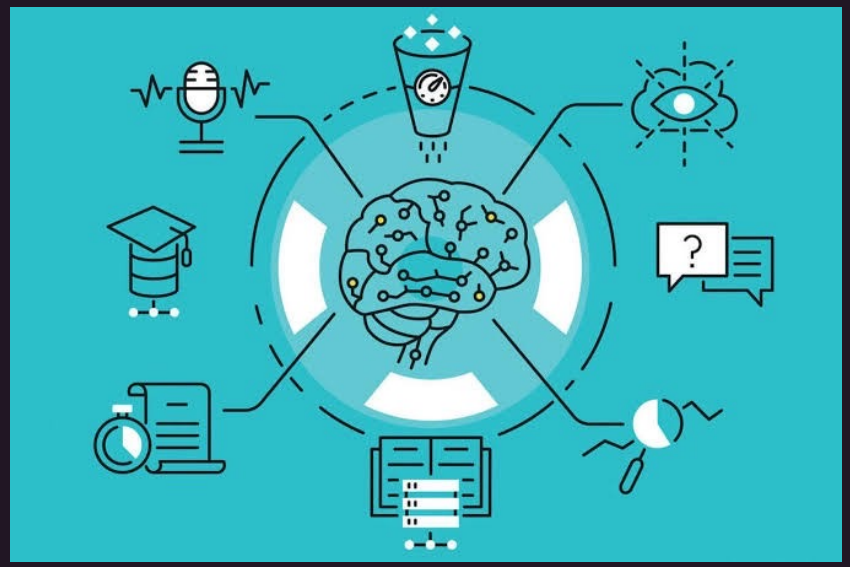
OpenAI releases Dall-E for generating images from text, and ChatGPT, a chat interface for GPT-3.5



Now

GenAI explosion

Widespread development of multimodal generative models across disciplines



Deep Generative Modeling: An Overview

AI, ML, DL, Generative Models

Artificial Intelligence

machines that mimic human intelligence and cognitive functions like problem-solving and reasoning

Machine Learning

algorithms that parse data and learn from it

Deep Learning

automates feature extraction, uses large data, “scalable” ML

Deep Generative Modeling

approaches that explicitly or implicitly model the distribution of inputs and outputs

Generative Modeling



Quantifying Uncertainty



Photo courtesy @teenybiscuit

Probability

- Limit of infinite number of trials, like a flip of a coin (Frequentist)
- Quantification of uncertainty (Bayesian)





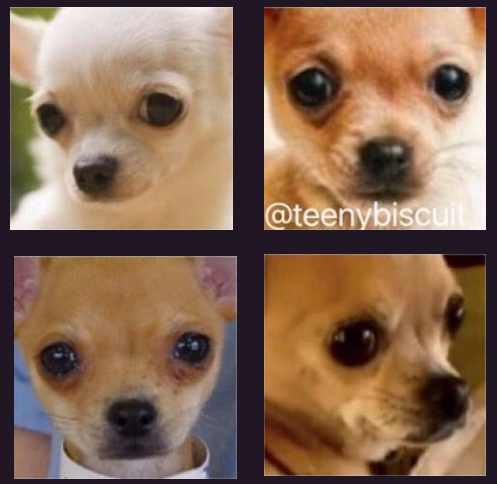
$$\mathrm{P}(\text{Concave Down}) = 60\% \qquad \mathrm{P}(\text{Concave Up}) = 40\%$$

$$\mathrm{P}(\text{Heads}) = \mathrm{P}(\text{Tails}) = 50\%$$

Example courtesy Chris Bishop

Statistical Generative Models

Given data samples $\mathbf{X}$ (images, floor plans, etc.), learn the probability distribution P_X



$\mathbf{X}$

$$P_X(\mathrm{X} = \quad) = 1$$

But if we have P_X, we can sample from P_X to generate new likely data samples e.g.

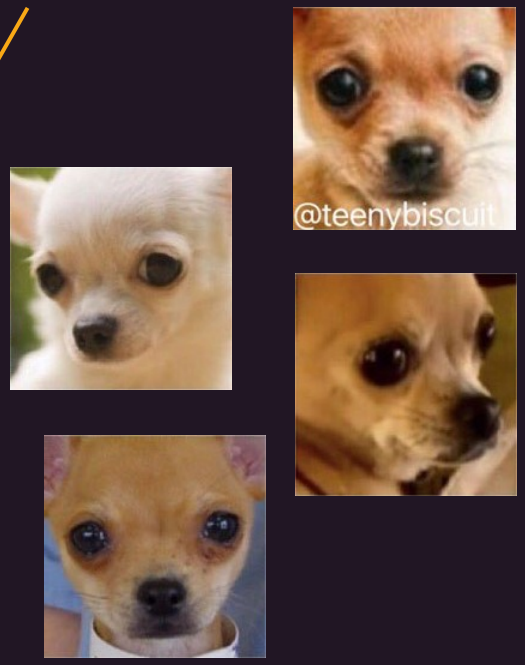


Generative because they can generate new samples by modeling the data distribution P_X

Revisiting the chihuahua vs muffin problem

$X =$  $\longrightarrow$ $Y =$ “muffin” ? $\Longrightarrow$ $P($ Y=“muffin” $|$ $X =$  $) = 0.78$





Decision Boundary

Discriminative Model

discriminates between which class Y a given image X belongs to, by finding the conditional distribution $P_{Y|X}$

Results from Chihuahua vs. Muffin at https://www.topbots.com/downloads/code/vision/chihuahua_vs_muffin/

Discriminative vs Generative Models

Discriminative Models

Directly model the conditional distribution over the variables from the data (here $P_{Y|X}$)

P(Y="muffin" | X = )

Generative Models

Model the joint distribution over the variables from the data (here P_{XY})

P(Y="muffin", X = )

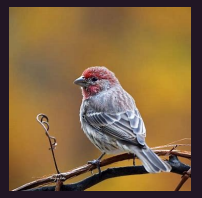
P(Y="chihuahua", X = )

Conditional Generation?

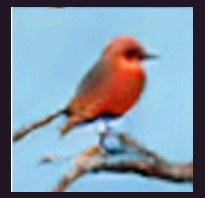
- Class conditional generation

$P(X = $  $| C = \text{“cat”})$

- Text conditional generation

$P(X = $  $| T = \text{“small bird with a red head”})$

- Image-Text conditional generation

$P(X = $  $| I = $  $, T = \text{“this bird with a red head and grey wings”})$

Types of Generative Models

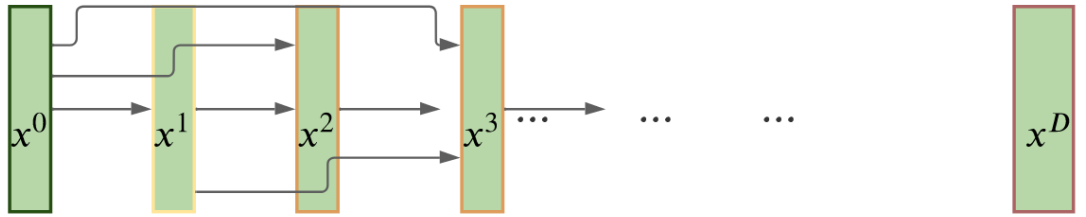
Types of Generative Models

Autoregressive Models

Factorize the joint distribution of a sample using the chain rule

$$p(\boldsymbol{x}_{1:\mathrm{T}}) = p(\boldsymbol{x}_1) \prod_{t=2}^{T} p(\boldsymbol{x}_\mathrm{t} | \boldsymbol{x}_{<\mathrm{t}})$$

Autoregressive model: Learn conditional of each variable given past



Density (evaluate $p(\boldsymbol{x})$) : Exact, fast
Sampling ($\boldsymbol{x} \sim p(\boldsymbol{x})$) : Slow
Training : MLE
Latents : None

Each conditional distribution is increasingly complex (owing to dependence on increasing number of variables), making computation intractable

Legend

$\boldsymbol{x}$	: observed data
$\boldsymbol{z}$	: latent code
$\boldsymbol{x}'$	: sample from the model
x^d	: d-th dimension of $\boldsymbol{x}$
R	: Real-valued output
$0/1$	: Binary output

Reproduced from Fig. 20.1, Probabilistic Machine Learning: Advanced Topics – Kevin Murphy

Types of Generative Models

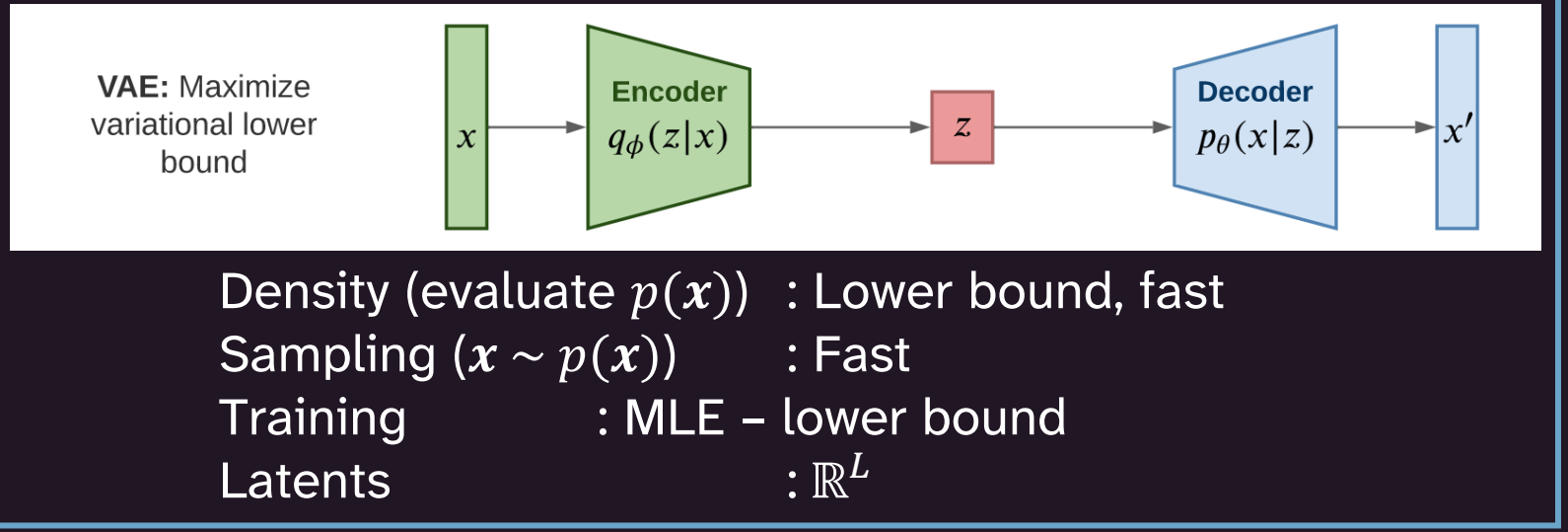
Variational AutoEncoders

Latent variable models, defining a joint distribution of the form

$$p_\theta(\boldsymbol{x}, \boldsymbol{z}) = p_\theta(\boldsymbol{z}) p_\theta(\boldsymbol{x}|\boldsymbol{z})$$

The posterior $p_\theta(\boldsymbol{z}|\boldsymbol{x})$ is intractable, so VAEs additionally define

$$q_\phi(\boldsymbol{z}|\boldsymbol{x}) = q\left(\boldsymbol{z}\middle|e_\phi(\boldsymbol{x})\right) \approx p_\theta(\boldsymbol{z}|\boldsymbol{x})$$



Density (evaluate $p(\boldsymbol{x})$) : Lower bound, fast
Sampling ($\boldsymbol{x} \sim p(\boldsymbol{x})$) : Fast
Training : MLE – lower bound
Latents : $\mathbb{R}^L$

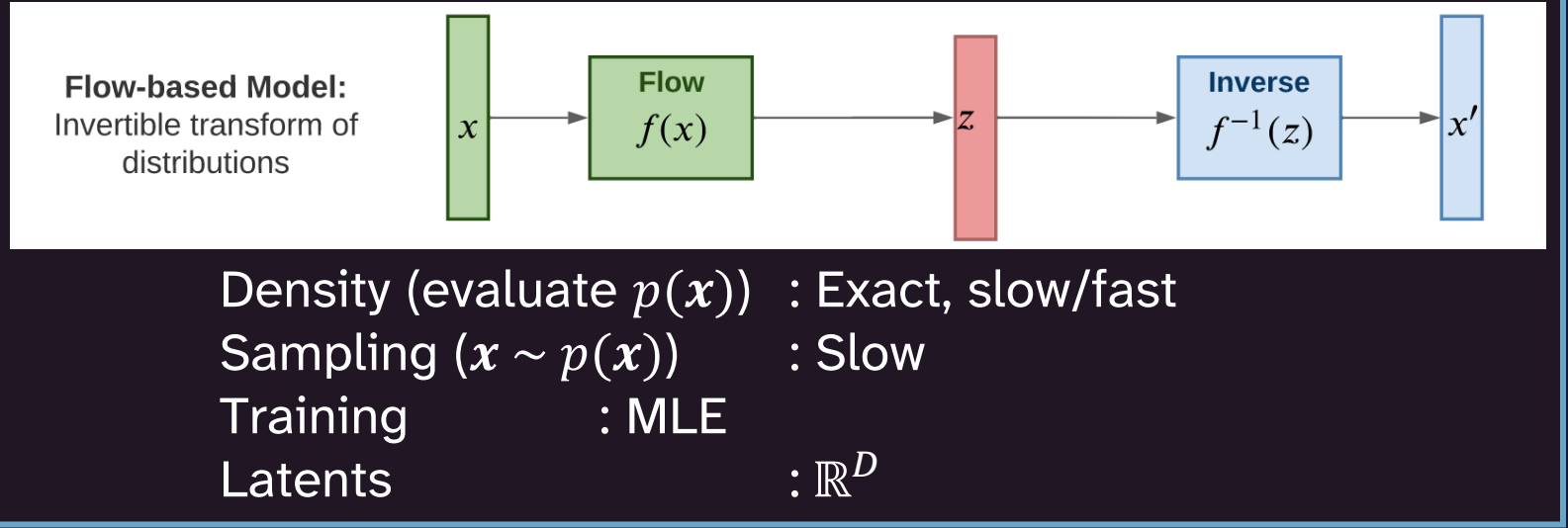
Legend

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$\boldsymbol{z}$	: latent code
$\boldsymbol{x}'$	: sample from the model
x^d	: d-th dimension of $\boldsymbol{x}$
R	: Real-valued output
$0/1$	: Binary output

Types of Generative Models

Flow-Based Models

Create complex distributions $p(\boldsymbol{x})$ from simple base distributions $p(\boldsymbol{z})$ by defining an invertible transform $f: \mathbb{R}^D \rightarrow \mathbb{R}^D$ from the data to latent space: $\boldsymbol{z} = f(\boldsymbol{x})$ and $\boldsymbol{x} = f^{-1}(\boldsymbol{z})$



Density (evaluate $p(\boldsymbol{x})$) : Exact, slow/fast
Sampling ($\boldsymbol{x} \sim p(\boldsymbol{x})$) : Slow
Training : MLE
Latents : $\mathbb{R}^D$

For computing density, we use change of variables:

$$p(\boldsymbol{x}) = p(\boldsymbol{z}) \left| \det \left(\frac{df(\boldsymbol{x})}{d\boldsymbol{x}} \right) \right|$$

Legend

$\boldsymbol{x}$	: observed data
$\boldsymbol{z}$	: latent code
$\boldsymbol{x}'$	: sample from the model
x^d	: d-th dimension of $\boldsymbol{x}$
R	: Real-valued output
$0/1$	: Binary output

Types of Generative Models

Diffusion Models

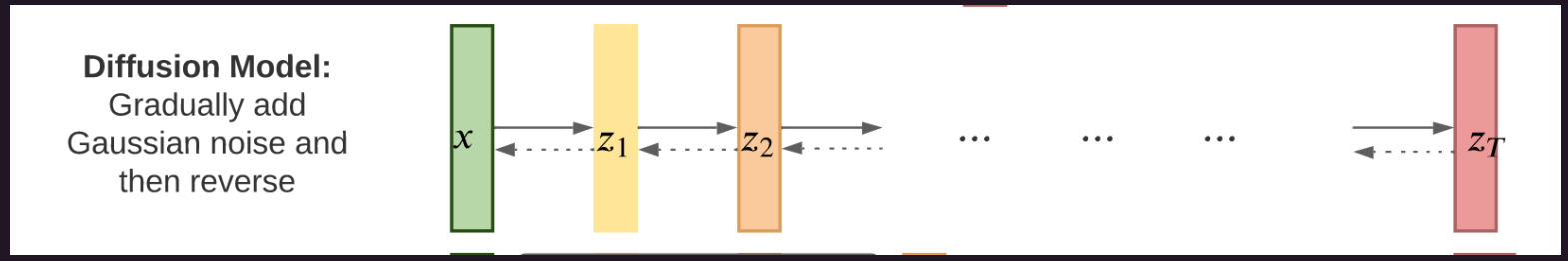
Define a Markov chain of noisy variables

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \mathbf{I}\right)$$

This turns data $\boldsymbol{x}_0$ into noise $\boldsymbol{x}_T \sim \mathcal{N}(0, \mathbf{I})$

Then learn a model to reverse this process

$$p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) = \mathcal{N}\left(\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t)\right)$$



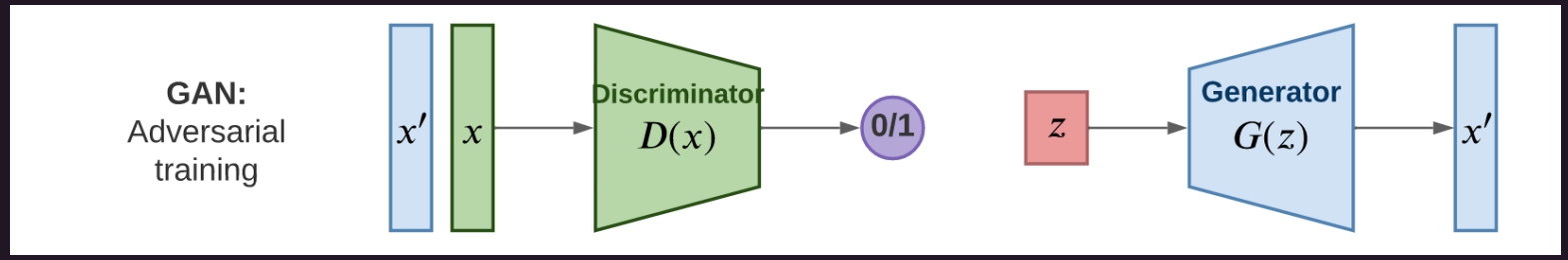
Density (evaluate $p(\boldsymbol{x})$) : Lower bound
Sampling ($\boldsymbol{x} \sim p(\boldsymbol{x})$) : Slow
Training : MLE – Lower bound
Latents : $\mathbb{R}^D$

Legend

$\boldsymbol{x}$	: observed data
$\boldsymbol{z}$	: latent code
$\boldsymbol{x}'$	: sample from the model
x^d	: d-th dimension of $\boldsymbol{x}$
R	: Real-valued output
$0/1$	: Binary output

Types of Generative Models

Generative Adversarial Networks



Density (evaluate $p(\boldsymbol{x})$) : N/A
Sampling ($\boldsymbol{x} \sim p(\boldsymbol{x})$) : Fast
Training : Min-max
Latents : $\mathbb{R}^L$

Implicit probabilistic models that do not provide a parametric specification of the distribution of X (like prescribed models do)

A discriminator $D(\cdot)$ distinguishes between real and synthetic data provided by a generator $G(\cdot)$ in the implicit form $p(\boldsymbol{x}|\boldsymbol{z}) = \delta(\boldsymbol{x} - G(\boldsymbol{z}))$ where $\delta(\cdot)$ is the Dirac delta function

Legend

$\boldsymbol{x}$	: observed data
$\boldsymbol{z}$	: latent code
$\boldsymbol{x}'$	: sample from the model
x^d	: d-th dimension of $\boldsymbol{x}$
R	: Real-valued output
$0/1$	: Binary output

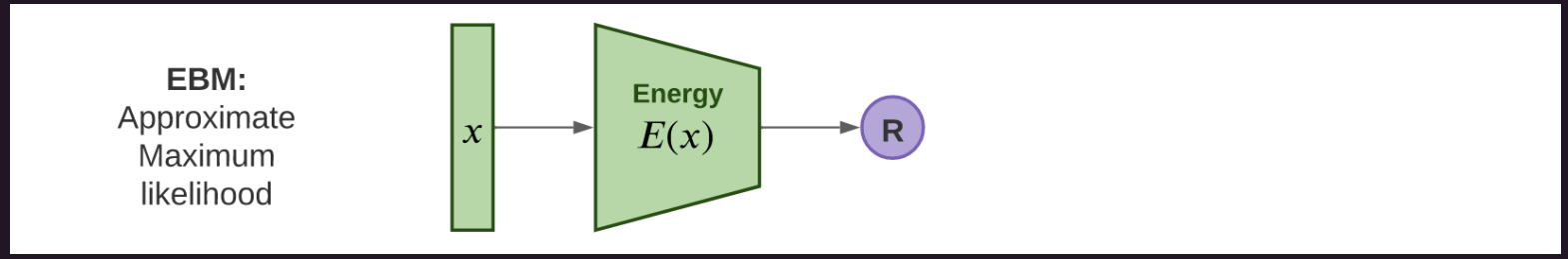
Reproduced from Fig. 20.1, Probabilistic Machine Learning: Advanced Topics – Kevin Murphy

Types of Generative Models

Energy Based Models

Define the data density using a Gibbs distribution:

$$p_\theta(\boldsymbol{x}) = \frac{\exp\left(-E_\theta(\boldsymbol{x})\right)}{Z_\theta}$$



Density (evaluate $p(\boldsymbol{x})$) : Approximate, Slow
Sampling ($\boldsymbol{x} \sim p(\boldsymbol{x})$) : Slow
Training : MLE – approximate
Latents : Optional

where

$E_\theta(\boldsymbol{x})$: energy function

$Z_\theta = \int \exp\left(-E_\theta(\boldsymbol{x})\right) d\boldsymbol{x}$: partition function

Legend

$\boldsymbol{x}$	: observed data
$\boldsymbol{z}$	: latent code
$\boldsymbol{x}'$	: sample from the model
x^d	: d-th dimension of $\boldsymbol{x}$
R	: Real-valued output
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Reproduced from Fig. 20.1, Probabilistic Machine Learning: Advanced Topics – Kevin Murphy

What this course is not specifically about

Training a language model from scratch

- Stanford CS336: Language Modeling from Scratch
  https://stanford-cs336.github.io/spring2025/
- Build a Large Language Model (From Scratch) - Sebastian Raschka
  https://github.com/rasbt/LLMs-from-scratch

Image/Video synthesis

- CMU 16-726: Learning-Based Image Synthesis
  https://learning-image-synthesis.github.io/sp25/lectures/

Course Logistics

Syllabus Highlights

- Course Website: https://genmod-3dbbb2.pages.ewi.tudelft.nl/

- Assessment
  - Digital Exam (60%)
  - Project Group Report (40%)

- Announcements on Brightspace, Discussions on Brightspace and Answers-EWI

- Reach out to any one of us with questions.

  Course email: DSAIT4030-EWI@tudelft.nl

Project

Goals:

- Hands-on experience with a generative modeling technique
- Deeper understanding of the process going from math to code
- Real-world understanding of engineering challenges

Process:

- Work in groups of 5/6 (split tasks ranging from data processing and loading, model implementation, and evaluation)
- Discuss plan with assigned TA
- Share findings in two presentations (mid-term and final)

Lectures

- Ask questions!
  - Interrupting the lecturer is encouraged
  - Others might also have the same questions
  - Asking questions through BS Discussions and Answers-EWI is also great
- When the instructor asks questions
  - Please answer and engage!
  - Think things through even if you don’t answer
- Interact and engage! We are all here to learn together and discover cool ideas